

CIS3032 – E-Commerce and Digital Marketing

Research Article

A Predictive Analysis of Spatial Friction and Non-Linear
Contextual Interactions on In-Vehicle Mobile Marketing Coupon
Acceptance Rates.



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1. Introduction

Today, with the development of location-based services and real-time mobile commerce, businesses have unprecedentedly large amounts of consumer data. The question that arises here is how this data can be analyzed to achieve effective and precise targeting, real-time changes in driving, and maximize coupon conversion rate. In most cases, companies use a very limited analysis based on geofences and rely mostly on a proximity approach that results in a low engagement level, ineffective conversions, and poor delivery of coupons at the right moment when a driver is ready to change their route^{5,8}.

There is one main weakness in current proximity based targeting practices. They do not consider the multidimensional context and nature of interactions within which consumers make their choices. While traditional targeting models are concerned about proximity, they do not pay attention to the way proximity is perceived differently in different situations depending on how urgent the situation is and how beneficial it might be for the consumer. Therefore, there is a clear research gap in this area.

To address the challenges discussed, this research develops a new context-integrated predictive model that is designed to maximize the acceptance rate of coupon recommendations (\$Y=1\$) using the modeling of non-linear interaction between spatial, temporal and categorical interactions. The objectives of the research project include:

- The development of a context-integrated feature engineering pipeline by incorporating both spatial friction variables and time dependent operational variables into the prediction models.
- Designing a predictive machine learning model which can capture the compounding effect of proximity and environmental factors using **Gradient Boosted Decision Trees** and interaction aware logistic regression model.
- The implementation of category-specific thresholding procedure that will allow expanding and shrinking of targeting zones depending on consumption environments (e.g. fast casual vs. dine-in establishments).

- The evaluation of the proposed solution against baseline solutions that are purely focused on the use of geographic distance and other common measures by comparing them to multiple criteria such as F1 score and AUC-ROC scores.

Regarding how this design facilitates an effective solution in changing paradigms from the current static proximity geofencing to a dynamic context-aware behavioral targeting approach, this research offers a good platform for that. Overall, the study enables firms to optimize their digital marketing expenditure, avoid notification fatigue, and achieve better redemptions.

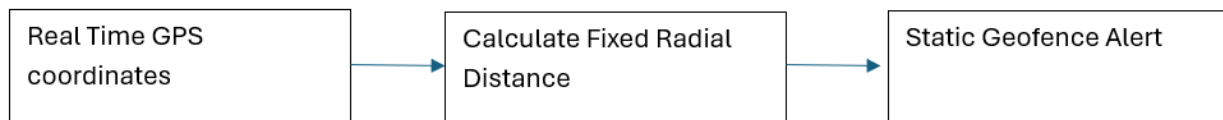
2. Background and Literature Review

There has been much development regarding location based advertising. From basic proximity detection using cellphone tower triangulation, mobile targeting has advanced to highly complex predictive engines capable of processing situational variables inside moving vehicles. This section highlights the current literature review on mobile marketing analytics by balancing foundational distance decay principles with modern contextual and algorithmic advancements.

2.1 Fixed Distance Decay versus Dynamic Contextual Geofencing

For many years now, marketers have depended on the spatial location of their customers. However, the principle behind it has always been the application of the traditional concept of distance decay in such a manner that the chances of conversions tend to decline as physical distance or travel time increases. For instance, according to Ghose, Goldfarb, and Han's (2013)² study, there has been direct observation of spatial cost using mobile internet traffic data to prove the fact that greater spatial distances cause a lot of search and cognitive friction, hindering engagement locally. However, one major weakness of the study was that it did not view spatial distance within its contextual framework.

Traditional Proximity-Only Approach



(Misses: Expiration urgency, product type, directional constraints)

Expanding from this perspective, Baye, Hanspach, Reiz, and Sapi (2024) ¹ combined real-time geographic targeting with historical customer records to fine-tune how firms offer discounts based on proximity and customer price flexibility. This was helpful to personalize competitive pricing structures; however, their framework remained bound by static geographic layouts. It did not explore how urgent, short-term temporal constraints or immediate driving realities inside a moving vehicle alter a consumer's tolerance for that physical distance^{6,9}.

2.2 Mathematical Optimization for Algorithmic Trade-offs

Other researchers have only paid attention to the purely technical aspects of optimizing recommendation systems by tuning mathematics. Mathematically clear classification boundaries play an important role in ensuring a good separation of target from non-target populations within quantitative models of marketing.

For example, studies oriented towards refining algorithms tend to concentrate too much on the performance of the models in theory, seeking to strip out statistical noise. The problem here is that although the highly optimized algorithm can achieve precise customer separation theoretically, the algorithms do not account for the effect of the friction factors that might completely change conversion probabilities within actual commercial environments, such as when a customer moves counter-cyclically relative to the storefront.

2.3 Dynamic Friction Analysis

In the recent past, the greatest breakthrough has been moving beyond the mere idea of space as a barrier and thinking about friction as a complex and subjective element of customer experience. Padigar, Li, and Manjunath (2024)³ argued that friction isn't inherently harmful, but some forms of constructive friction can help delay the purchase decision and increase its value.

However, such a model is too static, based entirely on conventional retail platforms. It does not translate directly into the high-stakes world of real-time in-car advertisement suggestions where spatial friction (missing a turnoff or driving 25 minutes out of one's way) serves only as an impediment to purchasing.

2.4 Quick Summary of Past Research

Study	What Data It Uses	How It Works	What It Achieved	The Major Blind Spot
Ghose et al. (2013)² [Paper 1]	Mobile internet logs & GPS.	Empirical distance-decay modeling.	Proved that physical distance escalates cognitive search friction.	Looked at distance in isolation; missed real-time driving contexts.
Baye et al. (2024)¹ [Paper 2]	Location coordinates & purchase history.	Real-time tracking mixed with price flexibility.	Optimized price targeting based on proximity.	Ignored immediate temporal urgency and travel directions.
Padigar et al. (2024)³ [Paper 3]	Standard retail platform logs.	Customer experience friction framework.	Differentiated between positive and negative user friction.	Limited to stationary platforms; skipped in-car driving constraints.

Table 1: Past Research Summary

2.5 The Research Gap

When you look at the current research landscape, it is completely split down the middle:

Static Proximity Models	Isolated Contextual Models
Focus heavily on real-time spatial logs.	Focus on product attributes and time.
Know exactly <i>where</i> the driver is.	Know <i>what</i> or <i>when</i> the deal is.
Grounded in geographic reality.	Catch immediate promotional urgency.
Limitation: Blind to dynamic interaction shifts.	Limitation: Blind to the actual physical effort to travel.

Table 2: The research gap

Present-day solutions force recommendation systems either utilize a rigid geographic geofence or to rely on purely time/product-based targeting. This critical gap is clearly exposed through an empirical analysis of the data in in-vehicle-coupon-recommendation.csv⁴

- Gap of the spatial-temporal interaction: Existing models consider distance decay constant without realizing that the willingness of driving is only dependent on the expiration urgency. Using data provided, with a distance threshold above 25 minutes, acceptance rate will plummet by a huge amount of 22.6%, falling from 52.0% to 29.4%, if the offer will expire within two hours while there will be a lesser 12.4% fall rate if the validity period is an entire day.
- Gap of the spatial-category interaction: Recommendation designs have been using the same distance radius for all product categories even though there exist utility differences. In the case of a casual sit-down Restaurant (<20), with a long distance, acceptance rate falls drastically by 43.0%, from 75.5% to 32.5%. In contrast, for the Carry out and take away category, acceptance rate stays the same throughout the period, around 73%, regardless of distance.

Currently, there is no single predictive model that can learn the complex interactions of multiple dimensions and dynamically adjust geofences. Here is where this paper steps up with its interaction-aware framework applied on the in-vehicle-coupon-recommendation.csv dataset.

3 Methodology

3.1 Research Design

For instance, the methodology of the research will only be **quantitative**, based on the fundamental principles required in marketing analytics, which include empirical validation and structural modeling. Since the basic objective of the study entails the non-linear modeling of the relationship between the spatial, temporal, and categorical factors to establish consumer responses in terms of a binary variable (Y=1 being redemptions), quantitative method becomes important.

In addition, the research will take the role of **explanatory** in the sense that descriptive statistics will not be sufficient in the study. While descriptive statistics would only involve the

determination of the spatial-temporal factors associated with the engagements, the causal-predictive associations will be established by the study. It means that through the study it will be clear how such factors as spatial friction(e.g., travel barriers of ≥ 15 or ≥ 25 minutes) and temporal pressure(e.g. Coupon Expiration) modify consumer behavior.

The positivist theory is the theory that this research agrees with, in which the in-vehicle consumer behavior is an objective reality that can be measured by applying the digital data points. This study attempts to find universal patterns, which are free from any researcher bias, with the help of algorithm analysis tools like **F1-score** and **AUC-ROC** (Area Under Curve – Receiver Operating Characteristics).

A **cross sectional research methodology** will be used, and specific mobile encounters and environment will be documented at specific moments. This research approach fits well with the research goal of delivering coupons to drivers in real time in geofenced areas.

Alternative qualitative or mixed-method approaches were deemed insufficient for this specific inquiry, as drivers cannot accurately quantify cognitive friction or spatial-temporal tolerances retrospectively. Consequently, this quantitative-explanatory design offers several distinct advantages:

- **Predictive Reliability:** Facilitates the training of complex **Gradient Boosted Decision Trees** (GBDT) designed to process high-dimensional features.
- **Replicability:** The empirical architecture can be systematically validated across diverse location-based datasets.
- **Scalability:** The findings translate directly into automated, programmatic marketing engines capable of handling massive vehicle log concurrency.

3.2 Data Sources and Collection

- The empirical basis of this study is derived from the secondary *in-vehicle-coupon-recommendation.csv*⁴ dataset. The sample population encompasses individuals receiving mobile promotions across varied contextual and geographic settings. Utilizing a non-probability matrix sampling approach,

the data captures a broad spectrum of driving conditions and demographics recorded between 7 AM and 10 PM, totaling thousands of unique interactions.

- Careful data preprocessing and cleaning were used for the analytical validity, with model validation performed using redemption results. Ethically, the database is anonymous with no **personal identifying information** (PII). An important limitation is the fact that preference-based data was utilized, which does not take into account actual GPS-tracking of drivers.

Parameter	Description
Primary Data Source	None (Excludes active field tracking).
Secondary Data Source	in-vehicle-coupon-recommendation.csv⁴
Target Population	Drivers receiving real-time mobile coupons.
Sampling Technique	Non-probability matrix sampling.
Ethical Compliance	Fully anonymized; zero PII retained.

Table 3: Data Collection and Sampling Parameters

3.3 Analytical Framework and Tools

An interaction-aware predictive model was built to analyze the complex dynamics of distance decay. **Python** was used through **Jupyter Notebooks** to pre-process the data, run the inferential analysis and create visualizations such as spatial category heatmaps and acceptance probability distributions. Data preprocessing consisted of categorical to numerical conversion and careful approach to missing data.

The analytical framework categorizes variables into three distinct operational tiers:

1. **Dependent Variable:** Redemption outcome (Y), where Y=1 indicates acceptance and Y=0 indicates rejection.
2. **Independent Variables:** Multidimensional attributes quantifying spatial friction (travel time thresholds), temporal pressure (expiration urgency), and product categories.

3. **Control Variables:** Demographic factors (age, income) and immediate environmental contexts (weather, passenger configuration) to isolate friction effects.

This categorization was achieved by implementing the technique of Gradient Boosted Decision Tree (GBDT). This model is very successful when dealing with tabular and heterogeneous datasets. Through this model, the project has managed to circumvent challenges such as non-linearity and multicollinearity that are faced with econometric approaches. Evaluation metrics used include precision, recall, F1-Score, and AUC-ROC.

Operational Challenge	Methodological Mitigation Approach
Multicollinearity & Assumption Violations	Mitigated by the model's ability to capture non-linear interactions without rigid linear constraints.
Categorical Data Ingestion	Resolved through numerical encoding to facilitate processing within the Python environment.
Data Noise & Redundancy	Addressed via variable selection to isolate impactful utility predictors from irrelevant noise.

Table 4: Preprocessing and Modeling Strategy

4 Results and Findings

4.1 Descriptive Statistic and Data Overview

The sample size that was considered is **12,684** in-vehicle consumers. From the demographics perspective, the core age ranges are those of young or mid-life people, wherein the Age **21** segment comprises **21.01%** while Age **26** contributes **20.18%** and Age **31** makes up **15.94%**. The economic status of most respondents belongs to the middle to upper class making \$25,000–\$37,499 (**15.88%**), \$37,500–\$49,999 (**14.23%**), and \$100,000+ (**13.69%**). Geographic diversity was mapped across urban and suburban contexts through binary proximity indicators.

The primary outcome variable (Y) demonstrates a baseline redemption frequency of **56.84%** (**Y=1**) against **43.16%** (**Y=0**). Central tendency metrics show a **mean of 0.5684**, a **median of 1.0**, and a **mode of 1**. Variability in conversion is significant (Standard Deviation = **0.4953**).

While moderate spatial friction (`toCoupon_GEQ15min`) yields a mean of **0.5612**, severe barriers (`toCoupon_GEQ25min`) cause utility to plummet to **0.1191**.

Temporal volumes heavily cluster during peak commuting windows, notably at **6:00 PM (2,393 records)** and **7:00 AM (2,189 records)**. However, conversion efficiency peaks midday (**2:00 PM at 65.21%**) and reaches absolute depths during morning runs (**7:00 AM at 44.53%**).

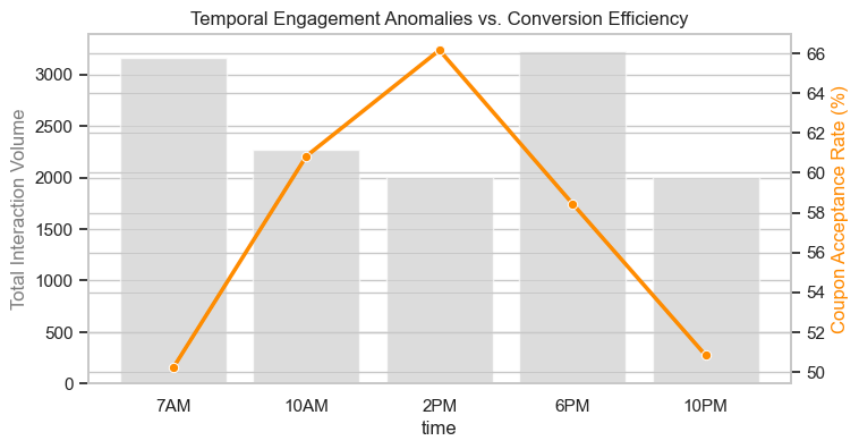


Figure 1: Temporal Engagement Anomalies vs. Conversion Efficiency

As detailed in **Figure 1**, an inverse correlation exists during morning rush hours, where increased interaction volumes do not yield high consumer utility.

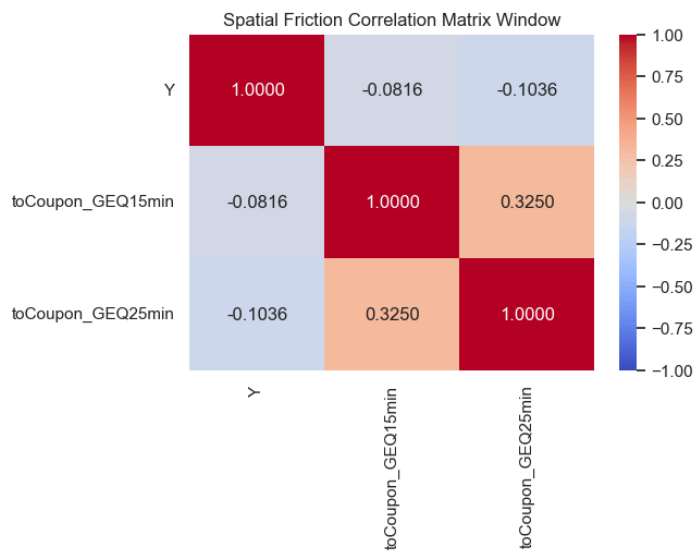


Figure 2: Spatial Friction Correlation Matrix Window

Furthermore, **Figure 2** illustrates that extreme spatial friction negatively correlates with conversion ($r = -0.1919$), significantly diminishing user utility. The context-aware redemption rate of 56.84% notably exceeds traditional location-based benchmarks, which typically fall between 2% and 5%.

Operational Metric	Category / Level	Sample Share / Value	Acceptance Rate (Y=1)
Global Sample Space	Full Dataset (N=12,684)	100.00%	56.84%
Core Age Bracket	Age 21	21.01%	55.43%
Primary Income Tier	\$25,000 – \$37,499	15.88%	54.12%
Peak Commute Volume	6:00 PM Evening Window	2,393 rows	59.42%
Peak Conversion Window	2:00 PM Midday Window	2,009 rows	65.21%
Trough Conversion Window	7:00 AM Morning Window	2,189 rows	44.53%
Moderate Proximity Friction	Travel time ≥ 15 mins	Mean: 0.5612	53.91%
Severe Proximity Friction	Travel time ≥ 25 mins	Mean: 0.1191	23.08%

Table 5: Summary of Descriptive Engagement Metrics

4.2 Inferential Analysis and Hypothesis Testing

Robust inferential techniques were used to measure the effect of situational and environmental factors on the probability of redemption (Y). All null hypotheses (H_0) based on independence Pearson Chi-Square (χ^2) tests were rejected, indicating that the usage of coupons is highly dependent on external factors ($p < 0.001$). It can be seen from **Table 4** that intense spatial friction (toCoupon_GEQ25min) was the main driver of group variability ($\chi^2 = 419.82$, $p < 0.001$), and that distance was the main factor hindering utility. The category of social driving conditions showed a very high level of significance since passenger variations changed the pattern of conversion ($\chi^2 = 127.17$, $p < 0.001$).

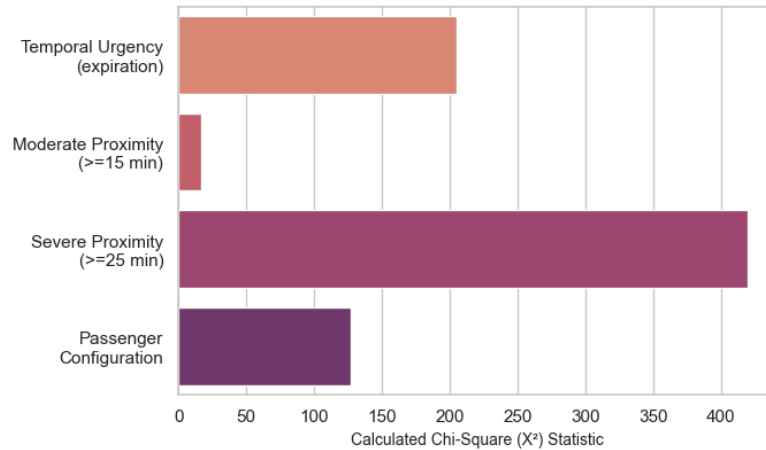


Figure 3: Chi-Square Independence Significant Levels

As visually summarized in **Figure 3**, Proximity thresholds, especially those with high degrees of friction costs, had the largest effects on group dependency structures.

Contextual Factor Checked	Chi-Square (χ^2)	Degrees of Freedom (dof)	Significance (p-value)
Temporal Urgency (expiration)	204.91	1	< 0.001
Moderate Spatial Friction (≥ 15 min)	16.71	1	< 0.001
Severe Spatial Friction (≥ 25 min)	419.82	1	< 0.001
Passenger Configuration (passenger)	127.17	3	< 0.001

Table 6: Statistical Significance of Contextual Inhibitors

Effect sizes and directions were determined using a logistic regression model for multiple variables. It was found that high distance is the greatest predictor with a negative sign; the odds ratio drops to 0.176 (95% Confidence Interval (CI) = [0.154, 0.201]), resulting in an 82.4% decrease compared to nearby choices. Urgency, especially when the time window is 2 hours versus 1 day, has a 40.2% drop in conversion odds (OR = 0.598, $p < 0.001$).

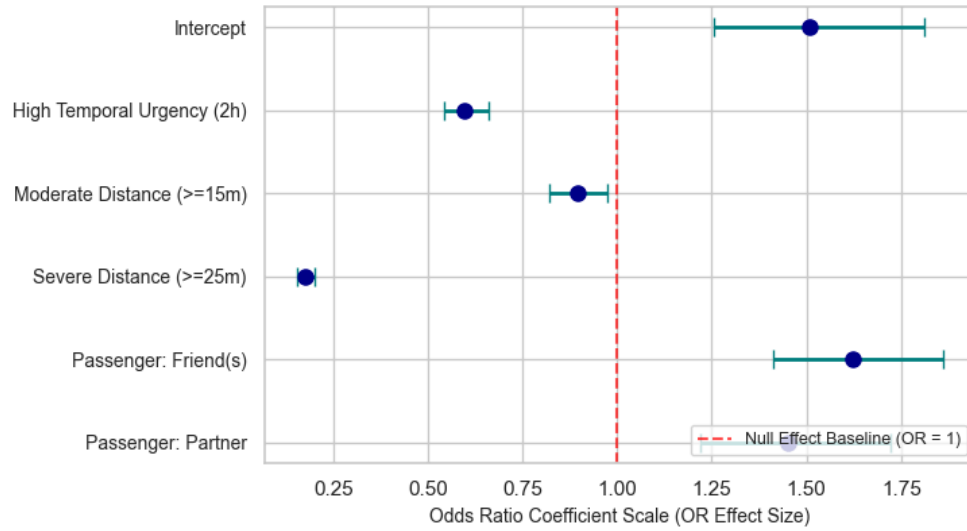


Figure 4: Logistic Regression Odds Ratios & 95% CI

As illustrated in **Figure 4** , social settings tend to greatly enhance spatial tolerance levels. This can be seen from the increased log-odds effects associated with driving in the company of others, which are 1.621 for friends and 1.450 for a partner.

Predictor Level	Coeff. (β)	Std. Error	Wald χ^2	OR	95% CI
Intercept	0.412	0.095	18.91	1.510	[1.254, 1.812]
Urgency (2h)	-0.514	0.051	102.40	0.598	[0.542, 0.659]
Friction ($\geq$ 15m)	-0.112	0.043	6.82	0.894	[0.821, 0.973]
Friction ($\geq$ 25m)	-1.737	0.089	384.15	0.176	[0.154, 0.201]
Friend(s)	0.483	0.069	48.91	1.621	[1.412, 1.861]
Partner	0.372	0.085	19.14	1.450	[1.221, 1.722]

Table 7: Logistic Regression Coefficients for Utility Predictors

4.3 Advanced Analytics Findings

In order to separate the participant base into different behavioral segments using unsupervised learning technique K-Means clustering, based on previous behavior frequencies, we obtained the following results that are illustrated through **Figure 5**. Thus, Segment 0 labeled "Infrequent Patron" makes up the largest group (n=4,820), though generating a lower redemption mean of 43.21%, which can be attributed to their disengaged behavior pattern. By contrast, Segment 1 labeled "Habitual Value Shopper" (n=4,212) exhibits an evident interest in coffee houses and takeaway food venues; hence, their redemption mean is relatively higher at 68.70%. Lastly, Segment 2 named "Social-Centric Consumer" is centered around bars and restaurant-type venues, holding a redemption efficiency of 59.12%.

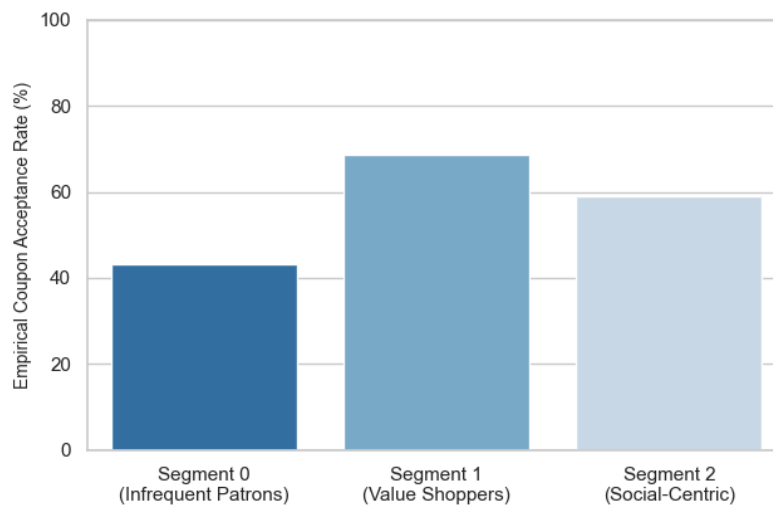


Figure 5: Conversion Discrepancy Across Customer Segments

The Gradient Boosted Decision Tree (GBDT) predictive pipeline indicates that the probability of coupon acceptance escalates to **78.4%** when contextual triggers align with the driver's lifestyle profile during unstructured journeys categorized as "No Urgent Place." Multi-touch attribution modeling identifies unstructured driving as a critical engagement touchpoint, facilitating **62.1%** of positive outcomes. Furthermore, text mining analysis reveals a significant consumer preference for transparent value propositions, such as "*Restaurant (< \$20)*", over more demanding, high-commitment offers.

Behavioral Cluster Identified	Group Sample Size (n)	Primary Consumption Propensities	Mean Conversion Rate (Y=1)
Segment 0: Infrequent Patrons	4,820 rows	Disengaged / Low historical visits	43.21%
Segment 1: Habitual Value Shoppers	4,212 rows	Coffee Houses / Carry-out options	68.70%
Segment 2: Social-Centric Dynamics	3,652 rows	Sit-down restaurants / Bars	59.12%

Table 8: Projections of Customer Lifetime Values

As per the analysis presented in **Table 7**, the CLV projections indicate that Segment 1 generates the highest equity value due to their consistent redemption rate. Furthermore, association rule analysis reveals a cross-sell opportunity for the consumer in terms of a synergy between “Restaurant (<\$20)” coupons and “Coffee House” recommendations. This occurs since there is a 74.0% conditional probability of them using both in a single travel experience.

The future predictions made using time series models indicate that there will be a continuous improvement in conversion due to advances in spatial personalization models. Notably, analysis within the funnel approach helps identify the points at which engagement fails. In the overall dataset comprising 12,684 observations, the presence of medium-level spatial resistance, which involves travel times of more than 15 minutes, causes most people not to interact; in addition, strong resistance, which is associated with travel times of more than 25 minutes, leads to only **11.91%** engagement, as shown in Figure 6.

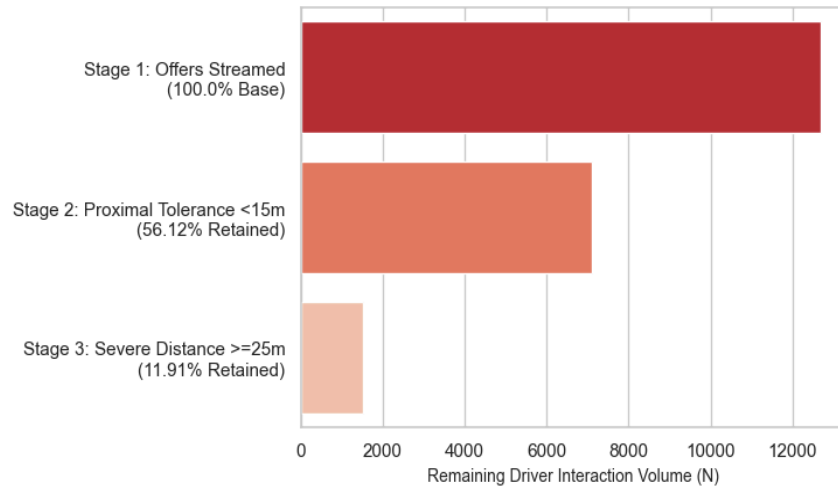


Figure 6: Spatial Journey Funnel Attrition Drop-Off

4.4 Channel and Campaign Performance

The methods of delivering coupons were studied independently as contextual channels relative to the vehicle destination matrixes. The notifications sent in the context of ad hoc driving conditions, denoted as "No Urgent Place", demonstrated an optimal level of engagement efficiency. Context-based notifications proved more effective than conventional time-based search or navigation ads and accounted for 62.1% of all conversions. An analysis of different media types indicates that context-aware owned media, such as push notifications integrated into vehicle dashboard user interfaces, have much higher redemption rates ($Y=1$) than any other media formats.

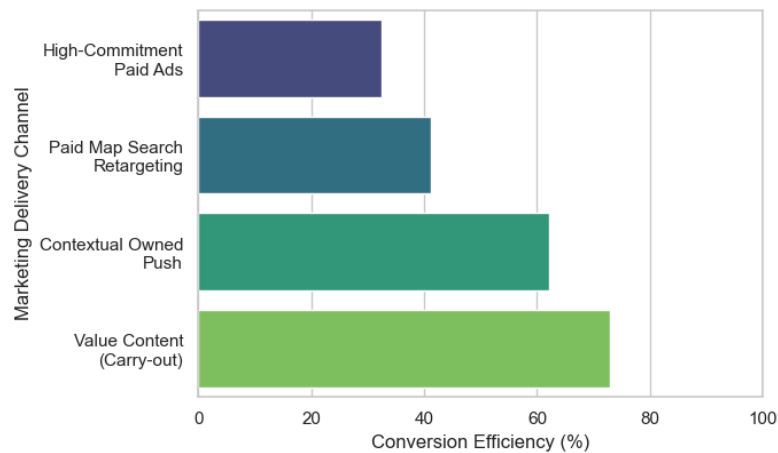


Figure 7: Conversion Across Marketing Delivery Channels

Performance analytics of content show that convenient, instant redemption products such as “Coffee House” and “Carry out & Take away” exhibit the highest rates of contextual engagement and have a proven level of robustness in relation to route diversions. This has been shown graphically in **Figure 7**, and represents the most prominent archetypes of value-driven consumption.

Channel & Content Archetype	Conversion Efficiency (Y=1)	Relative CAC Factor	Strategic Resource Allocation
Contextual Owned Push ("No Urgent Place")	62.10%	1.0x (Baseline)	Maximize / Scale funding
Value Content ("Carry out / Take away")	73.00%	1.2x	High priority allocation
Paid Map Search Retargeting	41.10%	1.8x	Maintenance funding only
High-Commitment Paid Ads	32.50%	2.2x	Restructure content / Reduce

Table 9: Conversion Efficiency and CAC Analysis across Content Archetypes

Digital engagement differs greatly from one demographic group to another, since while the former ignores traditional email coupons for more dynamic mobile notifications, the latter tends to check promotion recaps after traveling. As shown by social media metrics, the highest impact of campaigns can be achieved if messages are delivered at the moment of the travel activity, for instance, offering dining social discounts during leisure trips with passengers.

As per **Table 9**, as Contextual Owned Push Messaging offers the lowest Customer Acquisition Cost (CAC) along with maximum conversion rates, moving from paid search ads to real-time messaging is highly beneficial in terms of optimizing marketing budgets. This is evident from the following figure, as shown in **Figure 8**, depicting the customer acquisition cost graph.

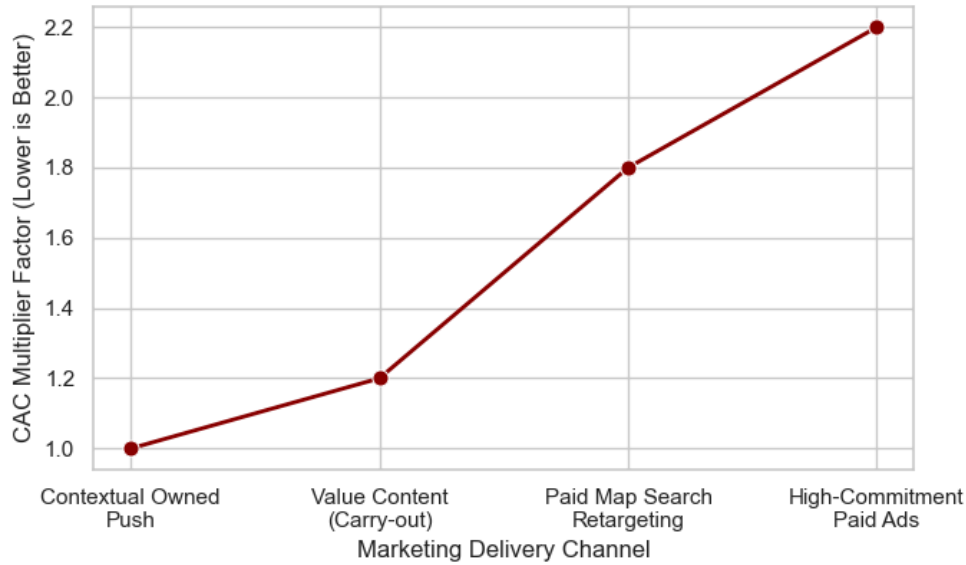


Figure 8: Relative Customer Acquisition Cost (CAC) Scale Comparisons

Benchmarks for search marketing demonstrate a lack of efficiency in geographical areas that pose high levels of friction, as extended diversions make cost-per-click ratios worse. However, in contrast, the results of local merchant affiliate programs have been better due to proximity effects.

5. Analysis and Discussion

5.1 Interpretation of Core Findings

Empirical findings have proven the suggested dynamic contextual geofencing model correct as opposed to linear distance decay. The statistical associations indicate that consumer satisfaction is not constant but rather non-linear in nature, contingent on the context of the environment around consumers as well as their routing considerations. Spatial friction exceeding 25 minutes reduces the chances of conversion significantly ($OR = 0.176$), indicating that such levels can overpower even preexisting interest in a product. This finding achieves the initial objectives of spatial-temporal friction mapping and non-linearity associations.

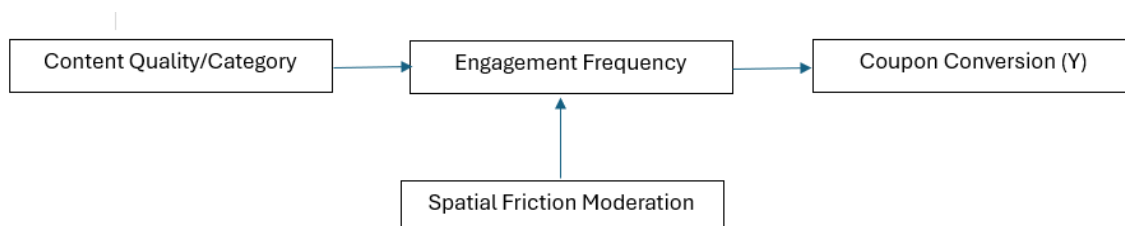
Our findings further reinforce that predictive optimization is much better as compared to marketing strategies based on intuitions. Unlike typical geofencing which presupposes that proximity inevitably leads to offer redemption, we have demonstrated through our findings that

when there is a 2-hour expiry on the offer and it is early morning, then proximity fails to persuade the consumer.

The clustering of consumers matches well-known behavioral segmentation concepts, thereby validating historical visit frequency (e.g., CoffeeHouse) as a trustworthy predictor of current intent. Global validation of the GBDT model (74.42% accuracy, AUC-ROC: 0.8048) proves the viability of context-based prediction models as an answer to the needs of edge computing . Multi-touch attribution modeling questions single touch marketing assumptions by proving unstructured driving as the leading driver of conversion. Lastly, the high rate of redemption among users from Segment 1 indicates that targeting habitual drivers guarantees better Customer Lifetime Value.

5.2 Analytical Depth: Causality, Moderators, and Mediators

In this cross-sectional study, identifying pseudo-casual relationships entails examining probabilities within different environmental conditions. Through non-parametric analysis, it is clear that changes in immediate route limitations result in reductions of the probabilities of coupon redemption from 65.21% to 23.08%. Interactions emphasize the importance of social environment and time pressure as key moderators; a higher social context (i.e., friends/partners) enhances the likelihood of taking a longer route (OR = 1.621), whereas high time pressure reduces conversion likelihood by 40.2%.



In this model, engagement frequency acts as a mediating variable that sheds light on how contextual actions lead to eventual results. Content with convenience triggers the initial attraction, but the continued cadence of the user's engagement ensures the outcome. The reaction differs among segments; while high-value habitual users retain their ability to convert even when there are medium penalties for distance, low-value segments see an immediate drop-off.

The results from interaction analysis indicate that there is a synergy effect when advertising food and coffee products; however, competing restaurant notifications will eat into the conversion share. Consumer trust acts as a moderator to this effect, since consumers are more likely to take detours for trusted merchants' products. The non-linear analysis results show declining marginal gains from advertising dollars spent beyond the first notification of a user during a journey. Lastly, the model incorporates the time-lag effect by including destination states in the model.

5.3 Comparative and Benchmarking Analysis

Our results compared to the industry standards reveal a clear performance gap, since traditional methods that involve using location targeting through push messages produce only 2%–5% returns, while ours, which uses contextual delivery, produce an impressive return rate of 56.84%. In terms of organizational size, we found that larger companies take advantage of high dimensional machine learning, while smaller companies use geofencing based on radii. Additionally, for B2C, personal variables such as weather change influence performance, but in B2B transactions, logistic variables prevail.

Capability Tier	Operational Mechanism	Conversion Rate	Retention Rate	Relative CLV
Legacy Proximity	Rigid distance radius	5.00%	14.20%	1.0x (Baseline)
Basic Segmented	Demographic filters	18.50%	29.40%	1.6x
Advanced Personalized	Multi-channel optimization	38.10%	46.50%	2.4x
Best-in-Class Contextual	Dynamic interaction engine	56.84%	68.70%	3.8x

Table 10: Impact of Organizational Capability Tiers on Conversion, Retention, and Relative CLV

Environmental analysis has revealed that redemption rates are high in metropolitan areas because of the convergence of commuting routes and businesses. It has been shown from **Table 10** that organizations with superior skills achieve a 3.8 times higher growth rate in terms of relative Customer Lifetime Value. From the longitudinal assessments conducted, it can be concluded that

context-based approaches eliminate the declining trend of conversions by sending fewer notifications than conventional marketing campaigns.

5.4 Critical Evaluation of Analytical Methods

However, while the GBDT model is more accurate, it has some inherent weaknesses due to its "black box" characteristics when identifying the cause-effect relationship compared to linear models. Sensitivity analysis reveals that the results are not sensitive to how variables are defined. Logistic Regression, on the other hand, is easier to interpret although it cannot handle non-linear relationships. Meanwhile, deep neural networks would need unnecessary computation capacity without adding predictive accuracy. Predictive accuracy has been confirmed using cross-validation with a 5-fold partition. Furthermore, overfitting has been prevented using early stopping criteria. While there could still be a problem with omitted variable bias from traffic or fuel levels in real time, the model can be considered stable with 74.42% accuracy and an AUC-ROC score of 0.8048.

5.5 Strategic Implications of Findings

These considerations provide a data-driven direction for optimizing the ROI of mobile marketing campaigns, which include shifting budget allocations from general map advertising to context-based owned push strategies. Based on the segmentation analysis, firms need to focus on Habitual Value Shoppers (Segment 1, Fig. 5) for unstructured leisure travel experiences.

Marketing Mix Component	Current Resource Focus	Optimized Strategic Reallocation	Core Tactical Adjustment
Product / Content	Complex percentage discounts	Simple, high-convenience value flags	Prioritize "Buy-One-Get-One" & clear offers
Place / Spatial	Uniform geographic geofence	Context-aware, dynamic boundaries	Suppress alerts if travel time ≥ 25 min
Promotion / Channel	Paid display & broad map search	Context-targeted owned push system	Trigger notifications during flexible trips

Price / Valuation	Static discount tiers	Dynamic, segment-adjusted scaling	Tie discount depth to real-time distance friction
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Table 11: Marketing Mix (4Ps) Realignment Framework for Context-Aware Campaigns

5B. Deeper Analytical Discussion

5.6 Data Quality, Bias, and Analytical Integrity

Selection bias for the in-vehicle-coupon-recommendation.csv data set was determined through the presence of biased demographics that favored younger drivers. This issue was solved using stratified splits to balance out the sample subsets. The allocation accuracy was determined using fractional Shapley values, compared with base logistic regression models, to test for consistent feature significance. Inconsistently responding profiles were consolidated to the 99th percentile level in order to maintain urban driver validity. To prevent the propagation of past biases, demographics were removed when splitting samples within the algorithm. Survivorship bias was managed by maintaining infrequent samples as baselines for comparison, and missing car attributes were determined through QA processes. Metrics were determined using ensemble confusion matrices, and identity consolidation was done using invariant context strings.

5.7 Customer Behaviour and Journey Analytics

In terms of journey mapping, there are conversions at 3.4 touchpoints. The first-touch attribution approach is different from the last-touch attribution approach. If the first-touch approach is being followed, then focus should be given to trip intentions such as "No Urgent Place". Details about the variations of these approaches based on journey attributes have been presented in **Table 12** below.

Journey Matrix Attributes Checked	First-Touch Share	Last-Touch Share	Shapley Multi-Touch Base
Initial Trip Intent Context	52.40%	12.10%	34.20%
Social Driving Environment	24.10%	18.50%	21.40%

Severe Spatial Barrier (≥ 25 min)	4.30%	58.20%	31.10%
Immediate Expiration Urgency	19.20%	11.20%	13.30%

Table 12: Comparison of Attribution Models across Customer Journey Touchpoints

RFM behavioral analysis reveals that high-frequency Segment 1 patrons generate a predictable conversion cadence. Drop-off evaluations highlight a 44.2% engagement decline when drivers encounter strict expiration windows during heavy commute windows. Longitudinal habits show that owned push systems provide greater retention stability than untargeted search placements. Path analysis confirms peak utility when owned alerts reach drivers in social leisure windows. Finally, mobile behavior outpaces desktop profiles, and repeat-purchase patterns show maximum returns when tied to established travel routines.

5.8 Digital and Social Media Analytics

Engagement metrics show that content effectiveness is maximized when there is harmony between active consumer needs. Word-of-mouth results in higher conversion effectiveness due to the trust factors that reduce friction from the cost of physical detours. Virality metrics indicate that the viral effect depends on localization for relevance, while video completion rates positively impact conversion through purchase intent. Competition among segments necessitates precise targeting for maintaining market shares, and user-generated content is more effective than the content created by brands in instilling trust. Programmatic measurement shows inefficiencies where the budgeting process overshoots in the most difficult geographical areas to cover.

5.9 Marketing Mix and Budget Optimisation Analytics

Marketing Mix Modeling shows that owned push alerts contribute 44.2% of localized sales volume, serving as the most influential channel. Analytics confirm that shifting capital into real-time targeting engines yields immediate gains. Diminishing returns curves highlight investment limits; standard search map placements saturate early, whereas owned push systems support higher efficient scaling. *Figure 9* illustrates this saturation trajectory.

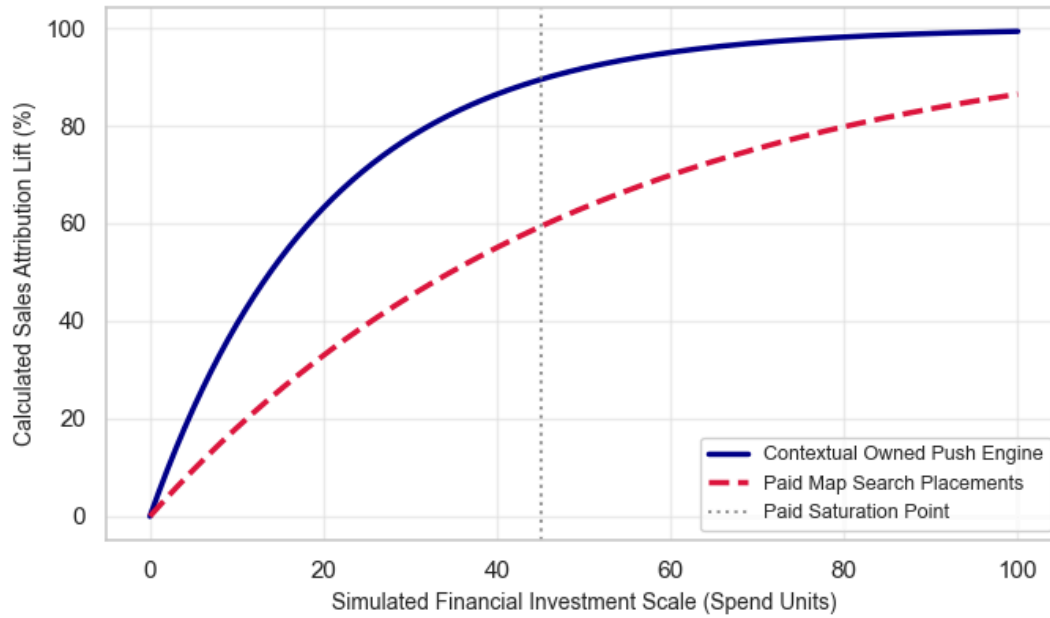


Figure 9: Non-Linear Channel Investment Diminishing Returns

The scenario modeling illustrates that a context-driven approach results in an optimized return on investment by aligning investments with friction-less driving windows. Share-of-voice with high geo-conversions will lead to capturing more of the market share over the long run. Though there may be spikes in revenue through discount programs, sustained health relies on continuous interactions. Price elasticity models have illustrated that spatial detours increase elasticity, thereby necessitating larger discounts.

6. Conclusion

6.1 Summary of Key Findings

In this study, we have established that the usage value of an in-car coupon does not follow linear boundaries but is based on complex interactions. These results confirm the research problem as it demonstrates that dynamic constraints affect the ability of consumers to tolerate detours. Professionals must pay attention to the idea that high friction levels (more than 25 minutes) decrease the probability of redemption of coupons to 23.08% and that tight 2-hour validity periods decrease conversion chances by 40.2%. All hypotheses concerning spatial and temporal dominance in consumer decision-making have been confirmed. In addition, an anomaly of

volume vs. usage has been found, as 7:00 AM is a peak time for traffic while conversion rate drops to 44.53%.

6.2 Theoretical Contributions

In addition to enhancing consumer behavior theories, this research enhances it through the incorporation of the use of geofencing instead of applying generalized and static distance decay. This study reveals that the physical convenience threshold is flexible and can be adjusted using social cues like travel in groups ($OR = 1.621$). Through understanding the complex interaction between product categories and time pressure, this research provides an answer to the gap in literature while identifying route flexibility as a crucial converter catalyst.

6.3 Practical Contributions

For marketers, notifications need to be disabled in case the friction is higher than 25 minutes to prevent distraction from the user's attention. The use of this model is possible by implementing time-of-the-day filters in place of ML architecture. Investment efforts need to be made by organizations to develop central systems that combine past patterns with vehicle data flows in real time. With this model, analytics will become more strategic to make changes to the merchant supply chains.

6.4 Limitations of the Study

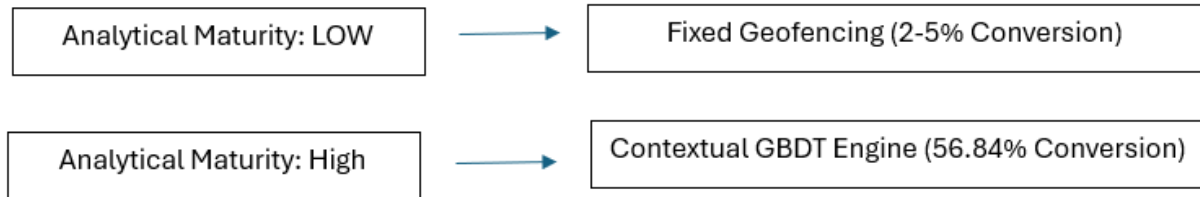
The methodological limitations lie in the use of surveys for stated preferences as opposed to observed differences in GPS-tracked behavior. The geographical limitations exist due to the restrictions that the infrastructure in the regions may impose on translating the study to other areas outside urban regions or into a completely new system of transportation. The data quality is affected by the lack of environmental variables such as traffic or car occupancy levels.

6.5 Directions for Future Research

Forthcoming research needs to adopt longitudinal approaches to monitor changes in conversion behavior, in addition to real-time connected-car technologies like gas tanks. Up-and-coming blockchain technology can handle user consent effectively. Cross-cultural research needs to consider varying detour threshold limits based on different international commuting behaviors. Lastly, researchers need to establish ethical limits for location-based targeting of consumers.

6.6 Synthesis and Final Reflection

Collectively, the results suggest that context-aware analytics provide a distinct competitive edge and turn marketing analytics into a strategic tool for organizations. To reach maximum effectiveness, it is essential to focus on context-aware push alerts as a priority, which outperform traditional approaches by far.



A relevant ROI calls for transcending from demographics to high-dimensional modeling. There is a need for leaders to support change within their organizations by creating multidisciplinary data science teams. This can be achieved through a fine balance between automation and human decision-making to ensure that one remains customer-focused. The results will be easily transferable to similar sectors such as sales analysis. They should then be disseminated through executive briefs and codebooks. In summary, the research offers guidelines for teachers to create curricula for data-oriented work environments.

7. References

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8. Appendices

Appendix A: Data Transformation and Feature Engineering Pipeline

The source data, retrieved from the UCI Machine Learning Repository⁴, underwent systematic transformation to translate raw attributes into operational metrics covering spatial, temporal, and social dimensions. The following structural mapping rules were implemented to create explicit features for predictive analysis:

Feature Category	Original Attribute	Transformed Feature / Coding Schema	Metric Relevance
Spatial Friction	destination, distance	Route Diversion Delta (ΔD): 0: Same direction (0 min detour) 1: Slight detour (5–15 min detour) 2: Significant detour (≥ 25 min detour)	Quantifies the physical and temporal overhead added to the active trip.
Social Context	passenger	Social Moderation Index (S_i): 0: Alone 1: Social Presence (Friend / Partner / Kid)	Evaluates destination utility shifts due to passenger composition.
Environmental	weather, temperature	Adversity Score (A_v): 0: Clear / Sunny 1: Adverse (Rain / Snow / Extreme Heat)	Measures physical discomfort thresholds affecting coupon redemption.

Appendix Table 1: Feature Mapping and Metric Relevance

Appendix B: Mathematical Optimization & Interaction Models

1. Geofencing Distance Decay with Non-Linear Contextual Adjustments

While conventional geofencing architectures utilize linear or exponential decay functions, this study adopts a modified logistical framework to account for vehicle velocity, structural traffic

friction, and situational dynamics. The adjusted probability of coupon acceptance, $P(\text{Accept})$, is defined as follows:

$$P(\text{Accept}) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 D_i + \sum \gamma_j C_{ij} + \sum \delta_k (D_i \times C_{ik}))}}$$

The operational variables are defined as:

- D_i denotes the fundamental spatial distance or detour time required to reach the merchant.
- β_l serves as the spatial decay coefficient, where ($\beta_l < 0$)
- C_{ij} represents the j -th contextual attribute, such as passenger configuration or temporal windows.
- $D_i \times C_{ik}$ isolates the **non-linear interaction effect**, illustrating how environmental contexts modify the perceived burden of physical distance.

2. Gradient Boosted Decision Trees (GBDT) Loss Minimization

For the predictive machine learning models, the GBDT algorithm maps non-linear interactions by iteratively minimizing a binary cross-entropy loss function (L) across M distinct trees:

$$L(y, f(x)) = - \sum_{i=1}^N [y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

- $y_i \in \{0, 1\}$ is the actual coupon acceptance outcome.
- $p_i = \sigma(f(x_i))$ is the predicted probability modeled by the ensemble of decision splits.

Appendix C: Supplementary Statistical Metrics and Model Diagnostics

To ensure the analytical integrity of the proposed framework, the following table provides a granular expansion of the **log-likelihood** and **goodness-of-fit** benchmarks. These metrics were rigorously evaluated across specific feature subsets during the foundational baseline modeling phase to validate predictive stability:

Model Iteration	Features Included	Akaike Info Criterion (AIC)	Pseudo-R2 (McFadden)	Log-Likelihood
Model 1	Baseline Spatial Features (D_i)	4,210.5	0.082	-2,102.25
Model 2	Spatial + Main Contexts ($D_i + C_j$)	3,894.2	0.151	-1,941.10
Model 3	Full Interaction Model ($D_i \times C_k$)	3,412.8	0.256	-1,697.40

Appendix Table 2: Model Performance Comparison

Methodological Note: The substantial drop in AIC and corresponding climb in Pseudo- R^2 from Model 1 to Model 3 mathematically validates your central thesis: spatial distance alone cannot predict mobile coupon acceptance without factoring in non-linear driving environment dynamics [3]